MMC HOST CONTROLLER

(1-BIT MODE)

Presented By

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Design Documentation

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1. Understanding the Big Picture

The MultiMediaCard (MMC) interface is a serial communication protocol used to connect an MMC memory card with a host controller. The host sends commands to the card, receives responses, and transfers data through dedicated data lines.

In this design, a 1-bit MMC Host Controller is implemented in SystemVerilog. The controller is responsible for:

- **Generating the MMC clock**
- **Sending command frames to the card**
- **Receiving card responses**
- **Handling data read/write**
- **Generating CRC for reliability**
- **Detecting timeouts**
- **Buffering data using FIFO**

The architecture separates the design into multiple modules, each responsible for a specific task.

The MMC Host Controller communicates with:

Host System

- Processor / controller that initiates commands

MMC Card

- Storage device connected via CMD and DAT0 line

SIGNAL OVERVIEW

Interface	Signals
System	clk, rst
Host Control	start, cmd_index, cmd_arg
Data Interface	read_en, write_en
MMC Bus	cmd, dat0
Status	busy, done, crc_error

The host sends commands such as:

- Card initialization
- Data read
- Data write

The controller manages the command transmission, response reception, and data transfer.

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2. Protocol Basics

MMC communication follows a command-response protocol.

Communication Lines

Signal	Description
MMC_CLK	Clock generated by host
CMD	Command line (bidirectional)
DAT0	Data line (1-bit mode)

Command Flow

HOST → CMD → CARD

CARD → RESPONSE → HOST

Steps:

1. Host sends command frame
2. Card processes command
3. Card sends response
4. Data transfer may begin

Data Transfer

Data transfers use DAT0.

Two modes exist:

Read

CARD → HOST

Write

HOST → CARD

Error Protection

MMC uses two CRC mechanisms:

CRC	Purpose
CRC7	Command frame protection
CRC16	Data transfer protection

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3. Key Features

This design implements the following major features.

MMC 1-bit Mode

Supports single data line communication.

Command Transmission

Commands are transmitted through the CMD line.

Response Handling

Card responses are received and validated.

CRC Protection

- **CRC7 for command frames**
- **CRC16 for data blocks**

Timeout Detection

Detects when a card does not respond.

FIFO Buffering

Stores data temporarily during transfers.

Clock Generation

MMC clock is generated using a divider.

Bidirectional Bus Control

CMD and DAT0 lines support tri-state control.

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4. Frame Format

MMC command frames contain 48 bits.

Command Frame Structure

Field	Bits	Description
Start Bit	1	Always 0
Transmission Bit	1	Host → Card
Command Index	6	Command number
Argument	32	Command parameter
CRC7	7	Error protection
End Bit	1	Always 1

Frame Representation

```
| Start | Tran | CmdIdx | Argument | CRC7 | End |
| 1bit  | 1bit | 6bit   | 32bit    | 7bit |1bit |
```

Total:

1 + 1 + 6 + 32 + 7 + 1 = 48 bits

Example command frame:

0 1 000000 00000000000000000000000000000000 CRC7 1

5. Basic Features Implemented in This Design

The following MMC host capabilities are implemented.

Command Controller

Builds command frame and starts transmission.

Command PHY

Handles physical transmission on CMD line.

Data Path

Handles data transfer on DAT0.

CRC Generators

Implements CRC7 and CRC16 algorithms.

Timeout Controller

Detects missing responses.

FIFO Buffer

Buffers data for read/write operations.

Clock Divider

Generates MMC clock from system clock.

Error Detection

Detects CRC mismatch in response.

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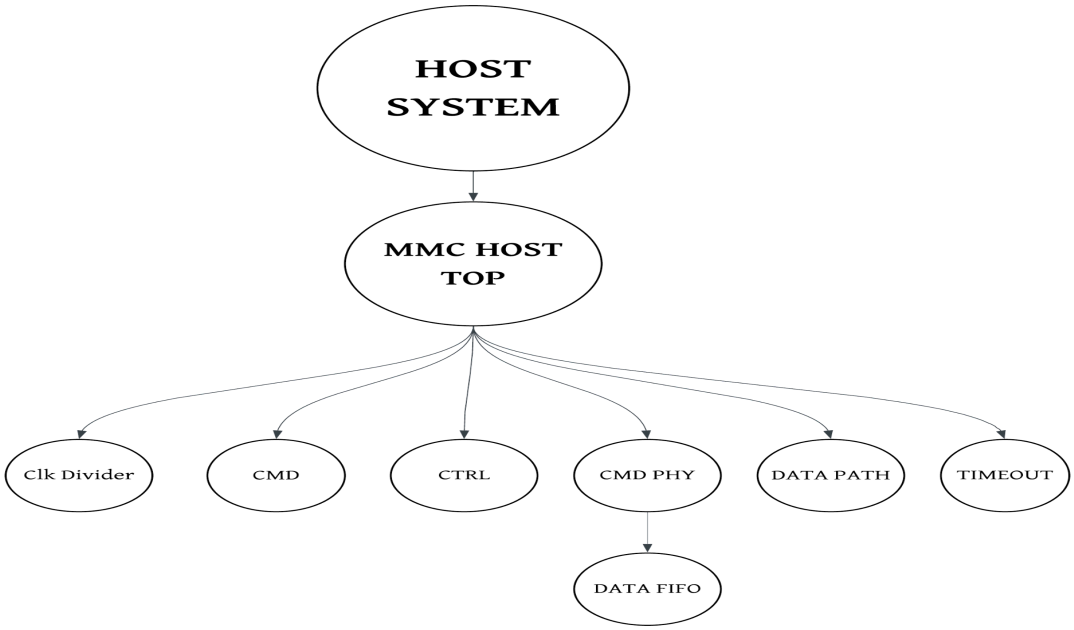
6. Micro-Architecture Components

The architecture is modular.



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MODULES

Module	Purpose
clock_divider	Generates MMC clock
mmc_cmd_ctrl	Controls command flow
mmc_cmd_phy	Handles CMD line
mmc_data_path	Handles DAT0 data transfer
crc7	Command CRC generator
crc16	Data CRC generator
data_fifo	Data buffering
timeout_ctrl	Timeout detection

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7. Block Diagram and Interface Description

7.1 MMC HOST TOP

Description

This is the top-level integration module connecting all blocks.

It manages:

- **Host interface**
- **MMC bus**
- **Internal modules**
- **Status signals**

Inputs

Signal	Source	Function
clk	System clock	Main clock
rst	System reset	Reset controller
start	Host	Start command
cmd_index	Host	Command number
cmd_arg	Host	Command argument
read_en	Host	Enable read
write_en	Host	Enable write
write_data	Host	Data to write

Outputs

Signal	Destination	Function
read_data	Host	Data read from card
busy	Host	Controller busy
done	Host	Operation complete
crc_error_flag	Host	CRC error detected

MMC Bus

Signal	Direction	Function
cmd	Bidirectional	Command line
dat0	Bidirectional	Data line

7.2 CLOCK DIVIDER

Purpose

Generates MMC clock from system clock.

Inputs

Signal	Source
clk	System clock
rst	Reset

Output

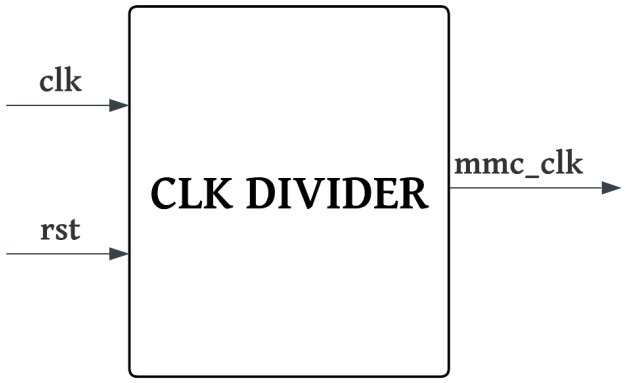
Signal	Destination
mmc_clk	All MMC modules

Function

Divides system clock using counter.

```
if(counter == DIV-1)
  mmc_clk toggles
```

BLOCK DIAGRAM



7.3 COMMAND CONTROLLER

Purpose

Creates command frame and controls command sequence.

Inputs

Signal	Source
clk	MMC clock
start	Host
cmd_index	Host
cmd_arg	Host
crc7	CRC7 generator
crc_ready	CRC ready
response_valid	CMD PHY
timeout	Timeout controller

Outputs

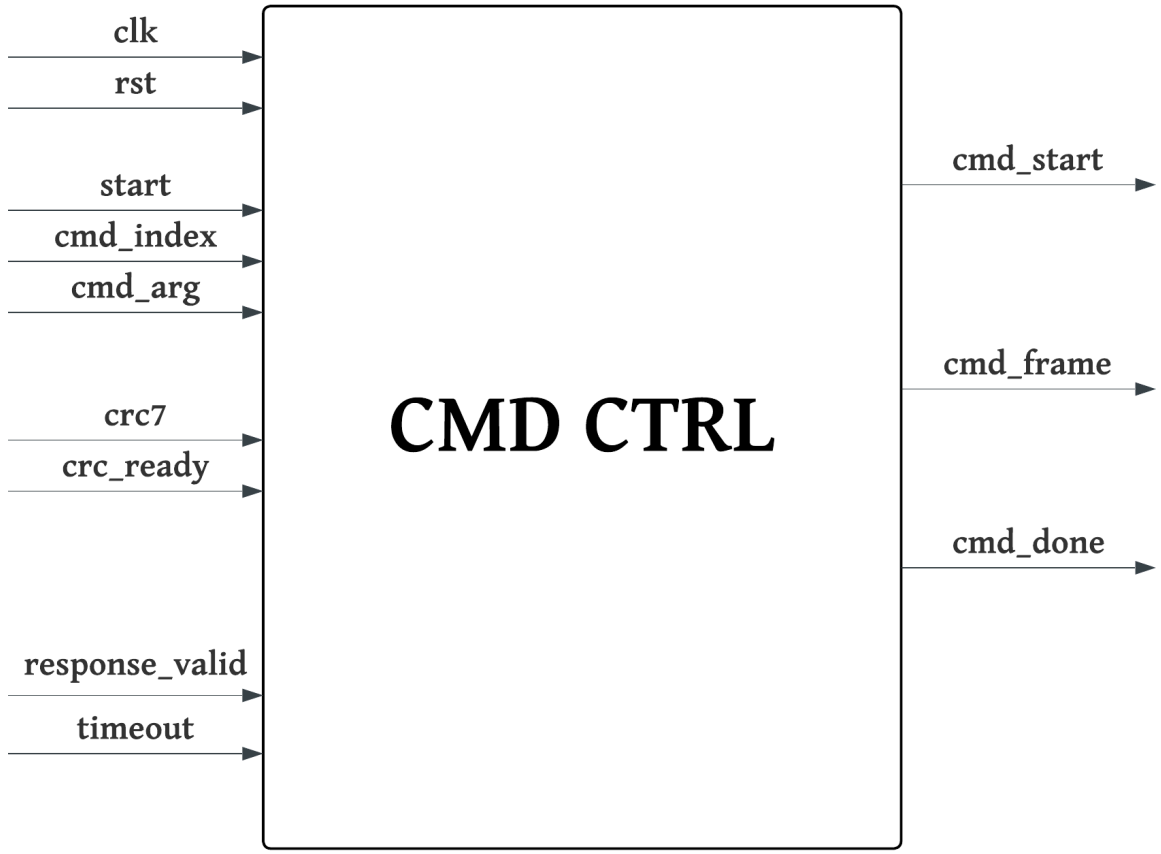
Signal	Destination
cmd_start	CMD PHY
cmd_frame	CMD PHY
cmd_done	Top module

Function

State machine:

IDLE
WAIT_CRC
SEND_CMD
WAIT_RESP
DONE
TIMEOUT

BLOCK DIAGRAM



7.4 COMMAND PHY

Purpose

Handles physical CMD line communication.

Inputs

Signal	Source
clk	MMC clock
cmd_start	CMD controller
cmd_frame	CMD controller
cmd_line	MMC card

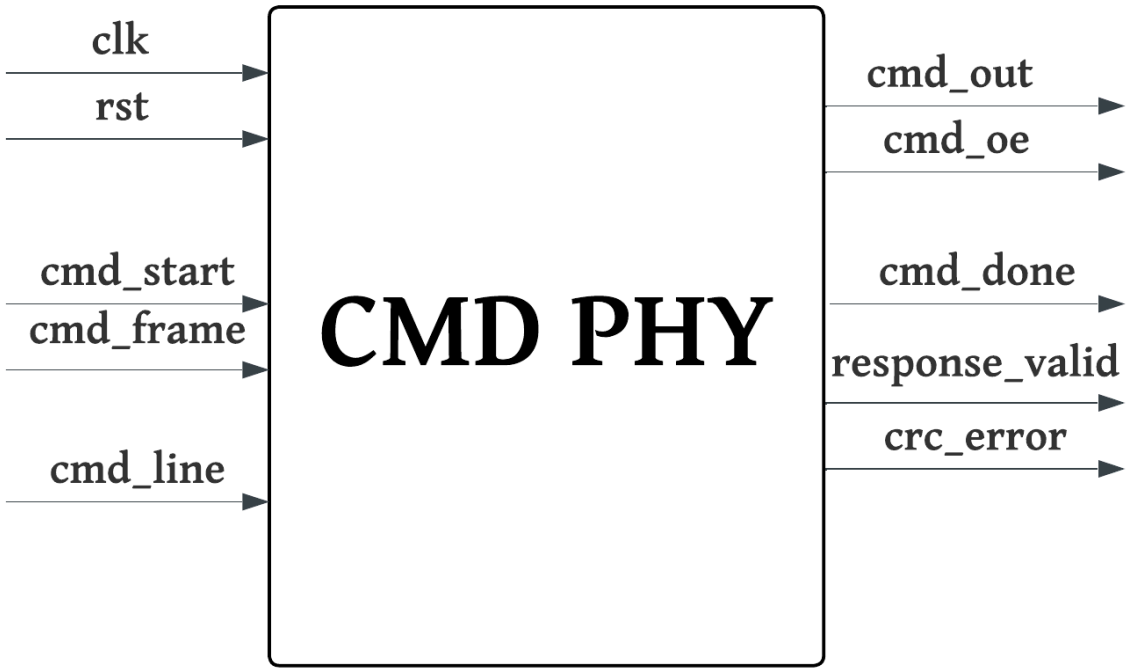
Outputs

Signal	Destination
cmd_out	CMD bus
cmd_oe	CMD bus
cmd_done	Controller
response_valid	Controller
crc_error	Top

Function

- Serializes command frame
- Sends bits on CMD line
- Receives response
- Verifies CRC

BLOCK DIAGRAM


clk
rst
cmd_start
cmd_frame
cmd_line
CMD PHY
cmd_out
cmd_oe
cmd_done
response_valid
crc_error

7.5 DATA PATH

Purpose

Handles data transfers via DAT0.

Inputs

Signal	Source
read_en	Host
write_en	Host
dat_line	MMC card
write_data	FIFO

Outputs

Signal	Destination
dat_out	DAT0
dat_oe	DAT0
read_data	FIFO
data_done	Controller

Modes

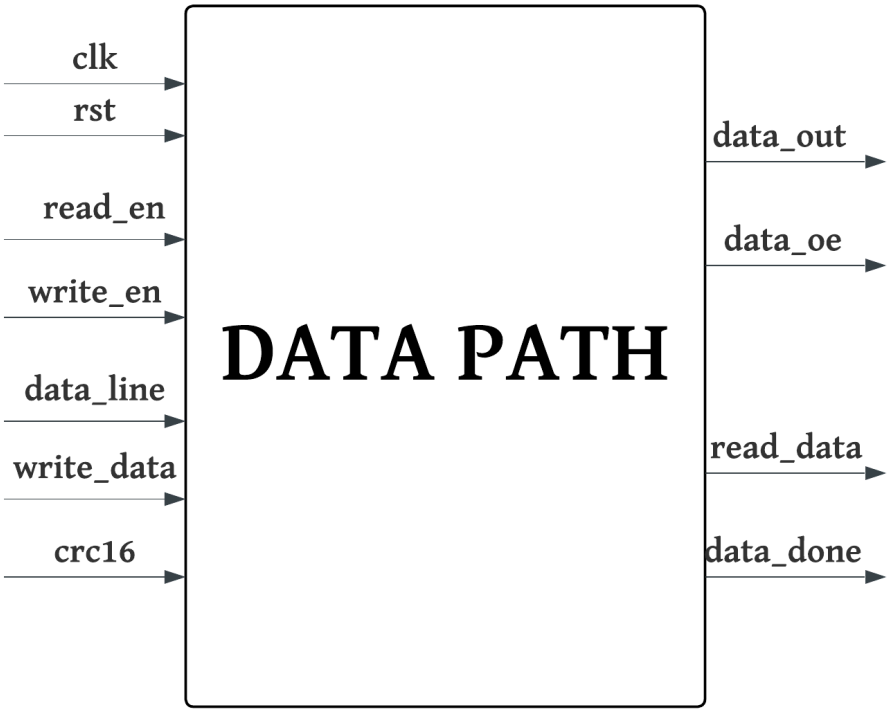
Write Mode

Host → MMC card

Read Mode

MMC card → Host

BLOCK DIAGRAM



7.6 CRC7 GENERATOR

Purpose

Generates CRC for command frames.

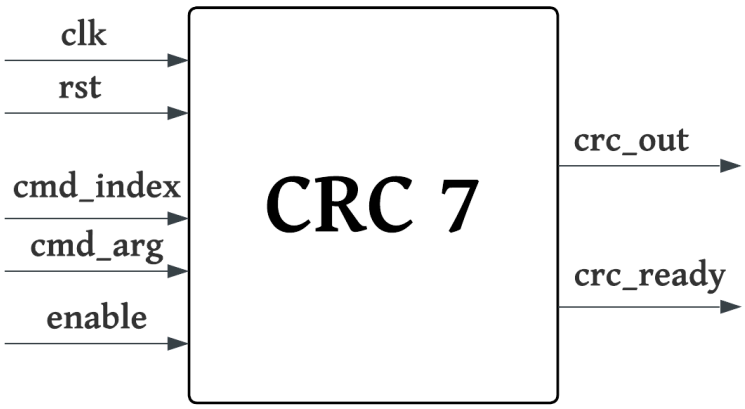
Inputs

Signal	Source
cmd_index	Command controller
cmd_arg	Command controller
enable	Start signal

Outputs

Signal	Destination
crc_out	CMD controller
crc_ready	CMD controller

BLOCK DIAGRAM



7.7 CRC16 GENERATOR

Purpose

Generates CRC for data transfers.

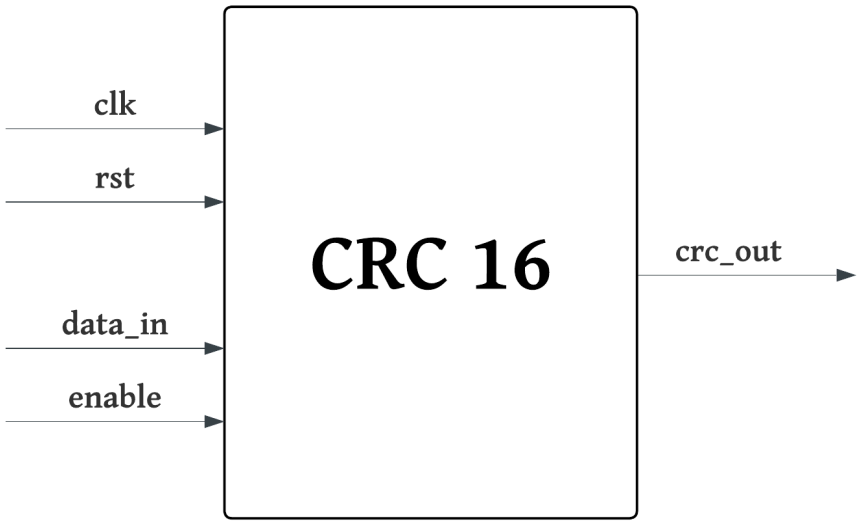
Inputs

Signal	Source
data_in	Data path
enable	write_en

Output

Signal	Destination
crc_out	Data path

BLOCK DIAGRAM



7.8 DATA FIFO

Purpose

Temporary data buffer.

Inputs

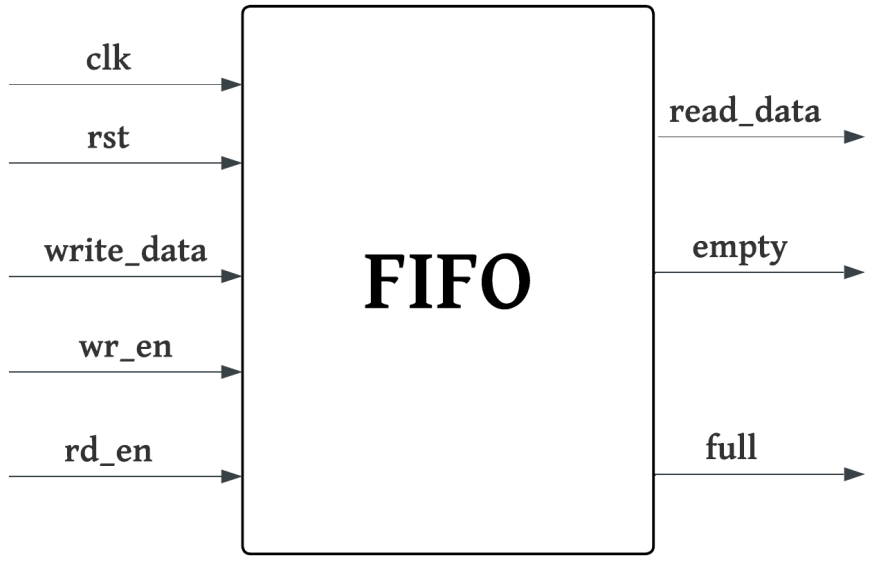
Signal	Source
write_data	Host
wr_en	Host
rd_en	Host

Outputs

Signal	Destination
read_data	Host
empty	Status
full	Status

FIFO Depth = 16 entries

BLOCK DIAGRAM


clk
rst
write_data
wr_en
rd_en
FIFO
read_data
empty
full

7.9 TIMEOUT CONTROLLER

Purpose

Detects response timeout.

Inputs

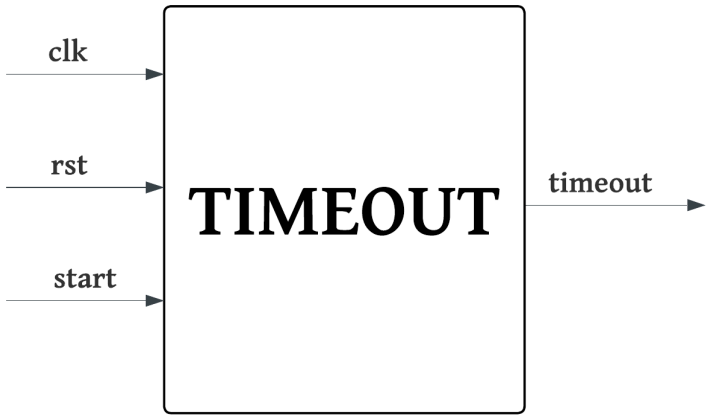
Signal	Source
clk	MMC clock
start	Command start

Output

Signal	Destination
timeout	CMD controller

Timeout limit = 100 cycles

BLOCK DIAGRAM



8. Implementation Details / Appendix

Design Language

SystemVerilog

Communication Mode

MMC 1-bit mode

Clock Generation

mmc_clk = system_clk / (2 × DIV)

FIFO Depth

DEPTH = 16

CRC Algorithms

CRC	Polynomial
CRC7	$x^7 + x^3 + 1$
CRC16	$x^{16} + x^{12} + x^5 + 1$

Command Frame Width

48 bits

Data Width

8 bits

Error Detection

- CRC mismatch
- Timeout detection